

Gene Pulser® Electroprotocol

Cell Type	Bacterial, gram positive	Molecules Electroporated	DNA: plasmid constructs (<i>Mycobacteria</i> / <i>E.coli</i> constructs)
Species Used	<i>Mycobacterium</i> , unspecified strain		

Before the Pulse

Cell Growth Medium	Tryptic soy broth (Difco)	Growth Phase at Harvest	O.D. (600) = mid to late log phase
		Pre-pulse Incubation	10% sucrose, 75mM phosphate buffer
Wash Solution	Distilled water		

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature	0 °C		
Electroporation Medium	10% sucrose, 75mM phosphate buffer	Cuvette Gap	0.2 cm
Cell Density	10 (10) to 10 (12) cells / ml	Voltage	2.4 kV
Volume of Cells	100 to 200 µl	Field Strength	12 kV/cm
DNA Concentration	1 to 5 µg DNA		
DNA Resuspension Buffer	TE buffer (10 mM Tris, 1 mM EDTA, pH 8.0)	Capacitor	25 µF
Volume of DNA	1/5 to 1/10 volume of cell suspension	Resistor	(Pulse Controller) 600 Ω
		Time Constant	5 to 10 msec

After the Pulse

Outgrowth Medium	Tryptic Soy Broth or Middlebrook 7H9 broth (Difco)
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Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	35 °C
Length of Incubation	overnight with shaking
Selection Method or Assay Used	antibiotic selection
Electroporation Efficiency	1 x 10 (3) to 10 (6)
Per Cent Survival	Not given

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Survey Number

077